Trust Systems: Canonical Trust, Certificates, and Consolidation

Extracted from a machine-checked development

Abstract

We develop the elementary theory of *trust systems*: dependency structures on claims, graded in a linearly ordered poset of check strengths. Each system carries a *canonical trust* assignment τ, characterised both as the greatest sound assignment of grades to claims and as the supremum of grades of derivations. Morphisms of trust systems transport canonical trust laxly. *Consolidations*—shape maps equipped with position pullbacks that cache grades—preserve τ on the nose. A circular-citation counterexample shows that well-foundedness of the premise relation is load-bearing; a shape-only consolidation counterexample shows that dropping checking steps degrades trust. All stated results have been machine-checked.

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1 Introduction

A recurring situation in formal and informal mathematical work is that a body of claims is justified by checking steps, each step resting on premises and receiving a grade recording how thoroughly it was checked. One wants a coherent global notion of how much each claim can be trusted, computed from the local data, and one wants to know how that global grade moves when claims are restated, imported, or consolidated across layers.

Trust systems make this precise. A trust system is a type of claims, a type of checking steps, and a grading of steps in a chain of check strengths. The *canonical trust* τ of a claim is the

widest-bottleneck value: the best derivation's weakest link. It may be defined either by well-founded recursion on the premise relation (fixpoint form) or as the supremum of derivation grades (max–min form); the two agree.

What this note claims. In ordinary mathematical language: every sound assignment of grades is bounded by τ, and τ itself is sound, so τ is the greatest sound assignment. Morphisms transport trust laxly ($\phi(\tau(x)) \leq \tau'(F(x))$). A consolidation with a genuine position map preserves trust exactly ($\tau'(F(x)) = \tau(x)$). Without well-foundedness of the premise relation, local certificates can certify circular trust that no derivation supports; without position pullbacks, restating claims alone can drop trust from the top grade to bottom.

2 Chains of check strengths

Definition 2.1 (Chain)**.** A **chain** is a type T equipped with a preorder $\leq$ such that

1. $\leq$ is a partial order (reflexive, transitive, antisymmetric);
2. $\leq$ is total: for all $a, b \in T$, either $a \leq b$ or $b \leq a$;
3. there is a bottom element $\mathrm{bot} \in T$ with $\mathrm{bot} \leq a$ for every $a \in T$;
4. $\leq$ is decidable.

We write bot for "no check performed".

Definition 2.2 (Meet and join)**.** On a chain T, define the binary meet and join by

$$\mathrm{cmin}(a, b) = \begin{cases} a & \text{if } a \leq b, \\ b & \text{otherwise,} \end{cases} \tag{1}$$

$$\mathrm{cmax}(a, b) = \begin{cases} b & \text{if } a \leq b, \\ a & \text{otherwise.} \end{cases} \tag{2}$$

Proposition 2.3. *For all $a, b \in T$:*

1. $\mathrm{cmin}(a, b) \leq a$ *and* $\mathrm{cmin}(a, b) \leq b$*;*
2. *if $c \leq a$ and $c \leq b$, then $c \leq \mathrm{cmin}(a, b)$;*
3. $a \leq \mathrm{cmax}(a, b)$ *and* $b \leq \mathrm{cmax}(a, b)$*;*
4. $\mathrm{cmax}(a, b)$ *equals a or equals b.*

Proof. By lean. □

We extend cmin and cmax to finite lists by right folds, writing $\mathrm{foldr}_{\mathrm{cmax}}(\ell, i)$ and $\mathrm{foldr}_{\mathrm{cmin}}(\ell, i)$ for the fold of a list ℓ with initial value i.

Proposition 2.4 (Fold properties)**.** *Let T be a chain.*

1. *If $a \in \ell$, then $a \leq \mathrm{foldr}_{\mathrm{cmax}}(\ell, i)$.*
2. $\mathrm{foldr}_{\mathrm{cmax}}(\ell, i)$ *equals i or belongs to ℓ (attainment).*
3. *If $a \in \ell$, then* $\mathrm{foldr}_{\mathrm{cmin}}(\ell, i) \leq a$*; also* $\mathrm{foldr}_{\mathrm{cmin}}(\ell, i) \leq i$*.*
4. *Pointwise domination of mapped values lifts to* cmin*-folds.*
5. *Every monotone map $\phi : T \to T'$ is lax for* cmin*-folds:* $\phi(\mathrm{foldr}_{\mathrm{cmin}}(\ell, i)) \leq \mathrm{foldr}_{\mathrm{cmin}}(\phi(\ell), \phi(i))$*.*

Proof. By lean. □

Remark 2.5. Attainment of finite joins (item (2) above) uses totality of the chain: in a general lattice a finite join need not be attained by any single element. All check-posets used in applications are finite chains.

3 Trust systems and canonical trust

Definition 3.1 (Trust system)**.** Let C be a type of claims and let T be a chain. A **trust system** S over C graded in T consists of

1. a type Step of checking steps;
2. a conclusion map $\mathrm{concl} : \mathrm{Step} \to C$;
3. a premise list $\mathrm{prems} : \mathrm{Step} \to \mathrm{List}(C)$;
4. a grade map $\mathrm{grade} : \mathrm{Step} \to T$;
5. for each claim x, a finite list $\mathrm{stepsInto}(x)$ of steps concluding x, such that $f \in \mathrm{stepsInto}(x)$ if and only if $\mathrm{concl}(f) = x$.

Definition 3.2 (Premise relation)**.** The **premise relation** $\mathrm{Prem}_S \subseteq C \times C$ is defined by

$$y \,\mathrm{Prem}_S\, x \quad \Longleftrightarrow \quad \exists\, f \in \mathrm{stepsInto}(x) \text{ with } y \in \mathrm{prems}(f).$$

Well-foundedness of Prem_S is the standing hypothesis for the certificate lemma below.

Definition 3.3 (Step grade)**.** Given an assignment $t : C \to T$, the **grade conferred by a step** f under t is the meet of the step's own grade with the assigned levels of its premises:

$$\mathrm{stepGrade}_S(t, f) \;=\; \mathrm{foldr}_{\mathrm{cmin}}\big(t(\mathrm{prems}(f)),\, \mathrm{grade}(f)\big). \tag{3}$$

Definition 3.4 (Canonical trust)**.** Assume Prem_S is well-founded. The **canonical trust** $\tau_S : C \to T$ is defined by well-founded recursion on Prem_S:

$$\tau_S(x) \;=\; \mathrm{foldr}_{\mathrm{cmax}}\Big(\{\, \mathrm{stepGrade}_S(\tau_S, f) \mid\; f \in \mathrm{stepsInto}(x) \,\},\, \mathrm{bot}\Big). \tag{4}$$

In words: $\tau_S(x)$ is the best step into x, valued at its step-grade under τ_S itself.

Definition 3.5 (Sound assignment)**.** An assignment $t : C \to T$ is **sound** if every claim trusted above bottom carries a node-local certificate: for every x with $t(x) \neq \mathrm{bot}$, there exists $f \in \mathrm{stepsInto}(x)$ such that

$$t(x) \;\leq\; \mathrm{stepGrade}_S(t, f).$$

Theorem 3.6 (Certificate lemma, upper bound)**.** *Assume* Prem_S *is well-founded. Every sound assignment is bounded by canonical trust: if* t *is sound, then*

$$t(x) \;\leq\; \tau_S(x) \qquad \textit{for every claim } x.$$

Proof. By lean. □

Theorem 3.7 (Certificate lemma, soundness of τ)**.** *Assume* Prem_S *is well-founded. Canonical trust is itself sound.*

Proof. By lean. □

Corollary 3.8 (Greatest sound assignment)**.** *Under well-foundedness of* Prem_S*, the assignment* τ_S *is the greatest sound assignment of grades to claims.*

Proof. By lean. □

Remark 3.9. The soundness of τ uses attainment of finite joins, hence totality of the chain. The upper bound (Theorem 3.6) would hold more generally.

4 Derivations and the max–min semantics

Definition 4.1 (Derivation)**.** A **derivation** of a claim x in a trust system S is a checking step f with $\mathrm{concl}(f) = x$, together with a derivation of each premise of f:

$$\frac{f : \mathrm{Step} \qquad (\forall\, y \in \mathrm{prems}(f).\ \mathrm{Deriv}_S(y))}{\mathrm{Deriv}_S(\mathrm{concl}(f))}\,. \tag{5}$$

Definition 4.2 (Derivation grade)**.** The **grade of a derivation** is the meet of every step-grade in the derivation tree (the weakest link): for a node with step f and sub-derivations of grades g_y,

$$\mathrm{derivGrade}\big(\mathrm{node}(f, \mathrm{sub})\big) \;=\; \mathrm{foldPrems}\big(\mathrm{prems}(f),\ y \mapsto \mathrm{derivGrade}(\mathrm{sub}_y),\ \mathrm{grade}(f)\big). \tag{6}$$

Theorem 4.3 (Fixpoint dominates derivations)**.** *Assume* Prem_S *is well-founded. For every derivation* d *of a claim* x*,*

$$\mathrm{derivGrade}(d) \;\le\; \tau_S(x).$$

Proof. By lean. □

Theorem 4.4 (Attainment by a derivation)**.** *Assume* Prem_S *is well-founded. If* $\tau_S(x) \neq \mathrm{bot}$*, then there exists a derivation* d *of* x *such that*

$$\tau_S(x) \;\le\; \mathrm{derivGrade}(d).$$

Proof. By lean. □

Corollary 4.5 (Agreement of fixpoint and max–min forms)**.** *Under well-foundedness, the fixpoint form of* τ *and the max-over-derivations of min-over-grades semantics coincide: whenever* $\tau_S(x) \neq \mathrm{bot}$*, the value* $\tau_S(x)$ *is attained by some derivation and bounds every derivation from above.*

Proof. By lean. □

5 Morphisms and transport of trust

Definition 5.1 (Morphism of trust systems)**.** Let S be a trust system over claims C graded in a chain T, and let S' be a trust system over claims C' graded in a chain T'. A **morphism** $M : S \to S'$ consists of

1. a map on claims $F : C \to C'$;
2. a map on steps $F_{\mathrm{Step}} : \mathrm{Step}_S \to \mathrm{Step}_{S'}$;

3. a map on grades $\phi : T \to T'$ that is monotone, satisfies $\phi(\mathrm{bot}) = \mathrm{bot}$, and is lax on step grades:
$$\phi(\mathrm{grade}_S(f)) \;\le\; \mathrm{grade}_{S'}(F_{\mathrm{Step}}(f));$$
4. premise preservation: $\mathrm{prems}_{S'}(F_{\mathrm{Step}}(f)) = F(\mathrm{prems}_S(f))$;
5. step-membership preservation: if $f \in \mathrm{stepsInto}_S(x)$, then $F_{\mathrm{Step}}(f) \in \mathrm{stepsInto}_{S'}(F(x))$.

Remark 5.2. Strictness of ϕ at bottom—"unchecked maps to unchecked"—is essential. Monotonicity alone fails when a source claim has no derivation but $\phi(\mathrm{bot})$ exceeds the target's bottom.

Theorem 5.3 (Lax transport of trust)**.** *Let $M : S \to S'$ be a morphism, and assume both premise relations are well-founded. Then for every claim x,*
$$\phi(\tau_S(x)) \;\le\; \tau_{S'}(F(x)). \tag{7}$$

Importing along a morphism never claims more than the image under ϕ of the source's canonical trust.

Proof. By lean. □

6 The circular-citation counterexample

Well-foundedness of the premise relation is load-bearing. We exhibit a two-claim cycle in which every node-local certificate passes, yet no derivation exists at all.

Definition 6.1 (Two-level chain)**.** Let **Two** $= \{\mathrm{lo}, \mathrm{hi}\}$ with $\mathrm{lo} \le \mathrm{hi}$ (and $\mathrm{lo} = \mathrm{bot}$).

Definition 6.2 (Circular system)**.** Let claims be booleans $\mathbb{B} = \{\mathsf{true}, \mathsf{false}\}$. Define a trust system cyc over $\mathbb{B}$ graded in **Two** by: the unique step into a claim x has single premise $\neg x$ and grade hi.

Proposition 6.3 (Constant-top is sound)**.** *The constant assignment $t(x) = \mathrm{hi}$ is sound on* cyc.

Proof. By lean. □

Proposition 6.4 (No derivations)**.** *No claim in* cyc *admits any derivation.*

Proof. By lean. □

Proposition 6.5 (Premise relation not well-founded)**.** *The premise relation of* cyc *is not well-founded.*

Proof. By lean. □

Remark 6.6. Circular citation is trust inflation invisible to any local check: well-foundedness is a global hypothesis and must be checked globally. The certificate lemma does not apply—as it must not.

7 Consolidation preserves trust

Consolidation restates claims (a shape map) and, when done with provenance, carries a position map: every checking step into a consolidated claim pulls back to a checking step into each source claim, with corresponding premises and a cached grade not exceeding the source grade.

Definition 7.1 (Consolidation)**.** A **consolidation** K from a trust system S over C to a trust system S' over C', both graded in the same chain T, consists of

1. a map on claims $F : C \to C'$;
2. a forward map on steps $F_{\mathrm{Step}} : \mathrm{Step}_S \to \mathrm{Step}_{S'}$ preserving premises (after applying F), lax on grades ($S.\mathrm{grade}(f) \leq S'.\mathrm{grade}(F_{\mathrm{Step}}(f))$), and preserving step-membership;
3. a backward map on positions: for each $x \in C$ and each step $f' \in \mathrm{stepsInto}_{S'}(F(x))$, a step $\mathrm{posn}(f') \in \mathrm{stepsInto}_S(x)$ such that

$$\mathrm{prems}_{S'}(f') \;=\; F\big(\mathrm{prems}_S(\mathrm{posn}(f'))\big)$$

and

$$S'.\mathrm{grade}(f') \;\leq\; S.\mathrm{grade}\big(\mathrm{posn}(f')\big).$$

The position map is indexed by the preimage x: when consolidation merges claims, every justification of the merged claim must pull back on *each* side.

Remark 7.2. The definition of consolidation never mentions τ. The condition is bisimulation-style and node-local; preservation of canonical trust is a theorem, not a definition.

Proposition 7.3 (Forward half is a morphism)**.** *Every consolidation determines a morphism of trust systems at $\phi = \mathrm{id}_T$.*

Proof. By lean. □

Lemma 7.4 (No trust inflation)**.** *Let K be a consolidation from S to S', and assume both premise relations are well-founded. Then for every claim x,*

$$\tau_{S'}\big(F(x)\big) \;\leq\; \tau_S(x).$$

Proof. By lean. □

Theorem 7.5 (Consolidation preservation)**.** *Let K be a consolidation from S to S', and assume both premise relations are well-founded. Then for every claim x,*

$$\tau_{S'}\big(F(x)\big) \;=\; \tau_S(x). \tag{8}$$

Proof. By lean. □

Remark 7.6. The forward inequality is lax transport at $\phi = \mathrm{id}$; the backward inequality is Lemma 7.4, using the position map to pull every target justification back to the source.

8 Composition of consolidations

Definition 8.1 (Identity consolidation)**.** The identity consolidation on a trust system S has identity maps on claims and steps, and identity position pullbacks, with exact (not merely lax) grade caches.

Definition 8.2 (Composition of consolidations)**.** If $K : S \to S'$ and $L : S' \to S''$ are consolidations (same chain T), their composite $K; L : S \to S''$ has

- shapes composed covariantly: $F_{K;L} = F_L \circ F_K$, $(F_{\mathrm{Step}})_{K;L} = (F_{\mathrm{Step}})_L \circ (F_{\mathrm{Step}})_K$;
- positions composed contravariantly: $\mathrm{posn}_{K;L}(f'') = \mathrm{posn}_K\big(\mathrm{posn}_L(f'')\big)$;
- cached grades chained by transitivity of $\leq$.

Theorem 8.3 (End-to-end preservation)**.** *If $K : S \to S'$ and $L : S' \to S''$ are consolidations and all three premise relations are well-founded, then for every claim x,*

$$\tau_{S''}\big((K; L)(x)\big) \;=\; \tau_S(x). \tag{9}$$

Moreover, the equation factors as the conjunction of the two layer-wise preservations.

Proof. By lean. □

Remark 8.4. Position maps compose as plain functions between step types—cofunctor composition—not via a comonadic duplicate. Trust is preserved end-to-end if and only if it is preserved layer by layer.

9 The shape-only counterexample

A consolidation that restates claims but drops checking steps is not a consolidation in the sense of Definition 7.1, and trust degrades.

Definition 9.1 (Ground system)**.** Let ground be the trust system over a singleton claim set with a single ground checking step graded hi (empty premises).

Definition 9.2 (Shape-only target)**.** Let layer4 be the trust system over a singleton claim set with *no* checking steps at all.

Proposition 9.3. *Both premise relations are well-founded (they are empty). Moreover*

$$\tau_{\mathrm{ground}}(*) \;=\; \mathrm{hi}, \qquad \tau_{\mathrm{layer4}}(*) \;=\; \mathrm{bot}.$$

Proof. By lean. □

Remark 9.4. No consolidation ground $\to$ layer4 can exist: there is no target step for the forward step map to hit. If one did, Theorem 7.5 would force $\tau_{\mathrm{layer4}}(*) = \mathrm{hi}$, contradicting the computation above. Shape-only restatement without position pullbacks is precisely the failure mode that degrades trust from top to bottom.

10 Discussion

What has been established. Canonical trust is the greatest sound assignment of check grades to claims, equivalently the max–min grade over the derivation tree, under well-foundedness of the premise relation. Morphisms transport trust laxly; consolidations with genuine position maps preserve it exactly and compose.

Corrections discovered by formalisation. Two hypotheses, invisible or underspecified in informal prose, are forced by the machine-checked development:

1. the check-poset must be a chain (for attainment of finite joins in the soundness of τ);
2. morphisms must send bottom to bottom (for transport on underivable claims).

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant. A mathematician reading this note need only verify that the English statements match the intended mathematics.

Acknowledgements

The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.